Sample science notes: Electricity

Original practice material for testing NCERT Tutor, not an official textbook.

Electric current is the rate of flow of electric charge. I = Q / t.

The SI unit of current is the ampere. An ammeter is connected in series.

Potential difference is work done per unit charge. V = W / Q.

The SI unit of potential difference is the volt.

Ohm’s law: at constant temperature, voltage equals current times resistance.

V = I R. Resistance is measured in ohms.

Example: 6 volts across 3 ohms produces a current of 2 amperes.

Sample science notes: Circuits and power

In a series circuit, the same current flows through every resistor.

Total series resistance is R = R1 + R2 + R3.

In a parallel circuit, each branch has the same potential difference.

For parallel resistors, 1/R = 1/R1 + 1/R2 + 1/R3.

Electric power is the rate of transfer of electrical energy.

P = V I. Using Ohm’s law, P = I squared R, and P = V squared / R.

The SI unit of power is the watt. Energy transferred is E = P t.

Example: a 60 watt lamp used for 2 hours consumes 120 watt-hours.

A fuse protects a circuit by melting when the current is too large.